Neuro 240 / Phys 240: Biological and Artificial Intelligence

Fall 2026 List of Potential Projects

Notes:

1. You can choose one of the projects below.
2. You can also design your own project, which requires approval from the course instructors:
   - The project should focus on the main theme of the class, i.e., AI and neuroscience
   - The project cannot be the same as a project that you are pursuing for another class
   - The project cannot be the same as a project that you are currently conducting in a lab
   - The project should be approximately comparable to the ones below in scope and difficulty

#	Project Title	Description, Hypotheses, Questions	References	Difficulty(0=easy, 10=hard)	Link / Data / Code
1	**Geometric constraints on prefix tuning**	Prefix tuning adapts a pretrained language model to a task by optimizing a continuous prefix, or "soft prompt," while keeping the model's weights frozen. Through causal self-attention, this prefix influences how the model processes later tokens in the sequence, affecting the output. Optimizing continuous representations directly removes the constraint that the prefix must correspond to discrete tokens, allowing more flexibility than a natural-language prompt. Typically, the prefix is optimized to maximize task performance, using gradients of the model's output cross-entropy loss. This project explores its use as a tool for mechanistic interpretability. Can we learn a soft prefix that selectively moves the internal representations of particular tokens in specified directions at a chosen layer? The central question is what geometric constraints are present in this form of task-adaptation. Are some directions in representation space more "promptable" than others? How do these constraints depend on the prefix length, the target layer, or the input? Can a prefix steer selected tokens without substantially changing the representations of others? How do these changes in internal representations affect the model's downstream behavior?	https://arxiv.org/abs/2101.00190	6	

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2	**Probing population codes through image synthesis**	Feature visualization technique reveals what individual units in a vision network respond to by optimizing an input image to maximize their activation. This project extends that idea to populations of neurons: can we generate a collection of images whose representations form a sphere (or some other geometric structure) around the representation of a seed image at a chosen layer? Representations of an object under different viewing conditions are often illustrated as a roughly ellipsoidal cloud in activation space. We can test this intuition by finding which image transformations are needed to move the population representation equally far in different directions. One approach is to combine a pretrained image generator with a frozen vision network, optimizing the generator's inputs to produce interpretable edits that move the seed image's representation toward specified targets. Are some directions easier to reach than others? Which correspond to changes in lighting, viewpoint, or background, and which require changing the object itself? How do these patterns depend on the layer and the distance from the original representation? By examining both the generated images and their activations, we can explore the geometric constraints on population codes and how they relate to meaningful visual changes.	https://arxiv.org/pdf/2311.17833	8	
3	**Computational role of delays in recurrent networks**	Conduction delays allow a recurrent network's dynamics to depend on its recent history, potentially expanding its computational capabilities without adding neurons. This project asks whether neural networks can exploit that additional temporal structure as a resource for memory and pattern generation. Can a recurrent network trained via backpropagation through time learn to use fixed delays as memory buffers? In a reservoir with frozen recurrent weights and a readout trained using FORCE, can delays produce dynamics that support richer or more accurate temporal outputs? Students could investigate how delay magnitudes and distributions affect performance on tasks requiring memory or the generation of complex temporal patterns. Are the benefits strongest in small		7	

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		networks or under constraints on connectivity? Do heterogeneous delays offer advantages over a single shared delay? Does the richer, potentially chaotic activity introduced by delays improve computation?			
4	**Can intermediate representations reveal the learning rule?**	Networks trained with different learning rules may achieve similar task performance, but do they arrive at similar internal representations? This project compares the representations learned through backpropagation, predictive coding, or equilibrium propagation, asking whether the geometry of intermediate activity contains a recognizable signature of the learning process. Do these methods produce different patterns of category separation, dimensionality, or sensitivity to irrelevant input variation? Can those differences reliably distinguish learning rules across independently trained networks, after accounting for architecture and task performance? A related direction examines how representations evolve during iterative inference. In a predictive-coding or energy-based network, hidden activity can change as the network settles toward a solution. How does this process reshape the representation of an input, particularly when it is noisy or ambiguous? Students can investigate whether iterative settling makes task-relevant information easier to read out, and whether the trajectory of neural activity during inference reveals differences that are absent from the final representation.	https://www.nature.com/articles/s41593-023-01514-1	6–8	

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5	**Can the geometry of internal LLM representations predict in-context generalization behavior?**	Large language models are known to be effective at in-context learning, where they are prompted with example demonstrations of some task that does not appear in their pretraining data and are then asked to perform the task on new samples. One simple type of in-context learning task is few-shot categorization. In this paradigm, the model is prompted with pairs of existing concepts and the desired response for each. For example, you could prompt the model with "dog → A, cat → A, bird → A" and "tree → B, flower → B, fern → B" and then ask it "bat → ?". From the given examples, the model must infer that the synthetic categories foo and bar refer to animals and plants. This is an example of an easy task, but can you construct harder ones by using less clearly related examples. Given such a task, can you predict the model's accuracy based on the geometry of its internal representations of each of the concepts (dog, car, bird, tree, flower, fern, bat)? A recent mathematical theory (paper linked) shows how the geometry of the internal model representations affects few-shot classification performance, assuming a very simple learning rule, but it is unclear whether this simple model can be applied to predict the behavior of LLMs. Can you apply this theory to link model representations and behavior in a quantitative way?	https://www.pnas.org/doi/10.1073/pnas.2200800119	5	
6	**Can you find the internal representation of an RL agent's goal and modify this representation to steer the agent?**	Beautiful experimental work in fruit flies (linked) has demonstrated a neural code for their goal direction (i.e., the spatial direction which they try to navigate toward). Optogenetically perturbing this neural code updates their apparent goal direction and causes flies to start navigating toward a new location. Do these ideas translate to artificial agents trained with RL to navigate toward simulated goal locations? First of all, can you decode the direction the agent is navigating toward from the internal representations of the policy network? Second, can you perturb the policy network's activity to "steer" the agent toward a new goal of your choosing?	https://www.nature.com/articles/s41586-023-07006-3 https://arxiv.org/abs/2310.08043	8	

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		Prior work (linked) achieved these goals for an RL agent navigating a simple 2D maze. Can you generalize this approach to a more complex agent navigating a 3D world (e.g., MuJoCo ant)?			
7	**Do vision-language models (VLMs) follow Weber's Law from psychophysics?**	Weber's Law is a famous result from psychophysics, which states that the smallest detectable change in a continuous quantity (e.g., the length of a line) is proportional to the value of that quantity. Does this also hold for VLMs? For the example of line length, you can construct sets of images containing two lines and ask the model whether the left or right line is longer. In this way, you can find the smallest difference in length that the model can reliably distinguish (the just-noticeable-difference or JND). Test whether the JND is approximately linearly proportional to the length of the lines shown. You can perform a similar analysis for other continuous features of visual objects (e.g., area, curvature, contrast, orientation). It may be best to use a small, non-reasoning VLM (e.g., PaliGemma) rather than a frontier model or at least prompt the model to avoid tool use to prevent it from directly measuring pixel-level length/area/etc. programmatically. Attached is an existing paper that performs similar experiments for *textual* representations of magnitude (e.g., textual descriptions of lengths in various units) for LLMs. The idea of this project is to extend this methodology to the richer visual domain.	https://en.wikipedia.org/wiki/Weber%E2%80%93Fechner_law https://arxiv.org/abs/2603.20642	3	

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8	**Do different AI models converge on the same representation?**	The "Platonic Representation Hypothesis" proposes that as models become larger and more capable, their representations converge across architectures and even across modalities. The evidence depends on how representational similarity is measured. Recompute alignment across model families and sizes (vision, language, or paired image–text data) using several metrics: mutual nearest neighbors, linear and kernel CKA, Procrustes distance, and RSA. Do the metrics agree on which models are most similar? How sensitive are the conclusions to dimensionality, sample size, and rare "outlier" dimensions with very large activations? Optionally, connect to Project 15: do models that converge with each other also align better with brain data? Compute: one 16 GB GPU; restrict to models of ≤3B parameters (or load larger ones in 8-bit).	Kornblith et al 2019: https://arxiv.org/abs/1905.00414 Huh et al 2024: https://arxiv.org/abs/2405.07987 Sun et al 2024: https://arxiv.org/abs/2402.17762	7	Code from Huh et al: https://github.com/minyoungg/platonic-rep
9	**How robust are claims about intrinsic dimension in language models?**	Several papers report that the intrinsic dimension of language-model representations rises and then falls across layers, and relate these phases to abstraction or linguistic processing. Replicate this finding with several estimators (TwoNN, maximum likelihood, participation ratio, local PCA) and test its sensitivity to context length, token position, and outlier dimensions. Then use the public Pythia training checkpoints to track how this profile emerges over training. Does it line up with known training events, such as the sudden appearance of induction heads? Compute: Pythia 70M–410M on one GPU; ID estimators run on CPU. Use 15–25 checkpoints and store them in /scratch (each 410M checkpoint is ~0.8 GB).	Valeriani et al 2023: https://arxiv.org/abs/2302.00294 Biderman et al 2023: https://arxiv.org/abs/2304.01373 Olsson et al 2022: https://arxiv.org/abs/2209.11895	6	Pythia models: https://huggingface.co/collections/EleutherAI/pythia-scaling-suite scikit-dimension: https://github.com/scikit-learn-contrib/scikit-dimension

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10	**Decodable vs. used: does a linear probe find what the model actually relies on?**	A linear probe can often read out a concept from a network's activations — whether a sentence is true, its sentiment, its grammatical number. But being decodable does not mean the network uses that information. For a few concepts, compare linear separability (with control tasks, so that the probe is not simply memorizing) against causal relevance: erase the concept with LEACE, or ablate along the probe direction, and measure the change in the model's behavior. When does "geometrically separable" imply "functionally relevant"? How does this relationship change across layers? This connects directly to how neuroscientists interpret decoding results from neural recordings.	Hewitt & Liang 2019: https://arxiv.org/abs/1909.03368 Elazar et al 2021: https://arxiv.org/abs/2006.00995 Marks & Tegmark 2023: https://arxiv.org/abs/2310.06824 Belrose et al 2023: https://arxiv.org/abs/2306.03819	7	Datasets from Marks & Tegmark: https://github.com/saprmarks/geometry-of-truth LEACE code: https://github.com/EleutherAI/concept-erasure
11	**Building models that generalize well.** **There are three different options:** **1. Weather modifications** **2. Day-night** **3. Real vs Cartoons/Sketches**	Neural networks are notoriously bad at generalizing to test data which is significantly different from train data. Recent efforts try to work across such shifts. Implement some recent works, and try to suggest modifications which might do well.	Musat et al 2021 — Robustness to rain Madan et al 2024	7	See links to datasets in Madan et al 2024
12	**The problem of parameters in linear systems**	Current deep convolutional neural networks (CNNs) are typically underdetermined. Why is it that they do not overfit? Compute condition numbers, Rademacher averages for underdetermined and overdetermined linear systems to assess robustness.	Poggio, Kur, Banburski. Double descent in the condition number Radhakrishnan et al 2020	6	—
13	**Sharpened and faded object boundaries**	It is widely known that CNNs are biased to textures rather than shapes. Taking a dataset with segmentation maps, sharpen or blur the edges in the training data. How does this impact the texture-loving nature of CNNs?	Geirhos et al 2019	7	image-net.org
14	**Impact of changing Transition Function in Deep RL**	(a) Use reinforcement learning to teach a network to play a video game like PACMAN.	GitHub: tychovdo/PacmanDQN YouTube: QilHGSYbjDQ	8	github.com/tychovdo/PacmanDQN

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		(b) The transition function defines the probabilities with which PACMAN ghosts move. How does the RL agent perform when the probabilities of ghost movements are different in testing than in training?	Bono et al 2024		
15	**Enforcing brain-like activations**	Recent works have trained linear models which take as input a CNN layer's activations and map them to neuronal activations collected from brain measurements. Reproduce these results and build on them.	Yamins et al 2013 Madan et al 2024	5	—
16	**Graphical humor**	Write an algorithm that will predict human judgments on whether an image is funny or not (or quantitative values on how funny an image is).	Veedant et al 2024	10	Link to data and comments
17	**Visual illusions**	Are current computer vision systems susceptible to human visual illusions? How do CNNs see these images? Can we create more such images automatically?	Kreiman. The phenomenology of seeing Ullman 2024 Lotter et al 2020 Williams et al 2018	8	Link to dataset
18	**Working memory**	Create a model that can solve a variety of delay match-to-sample working memory tasks.	Miller. Working memory 2.0 Xiao et al 2023 Yang et al 2019	9	Link to discussion and ideas
19	**Turing project**	Test state-of-the-art algorithms as human imitators.	Zhang et al. Human or machine? Turing tests for vision and language	4	Link to data
20	**Interpretability in neural networks**	Evaluate the extent to which unit activations are “interpretable.”	Olah et al 2021 Bardon et al 2022	5	—
21	**Which AI said that?**	A lot of work has explored whether it is possible to detect text produced by large language models (e.g., the ones underlying ChatGPT). A related, less well-explored idea is large language model attribution: assuming that a piece of text was produced by AI, which AI was it? GPT-2, GPT-3, GPT-4, Llama, Claude, Bard, Falcon?	Uchendu et al 2020 Munir et al 2021	4	GitHub repo from Uchendu et al paper

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		Build on existing work (see references) to implement a classifier that can distinguish between text produced by two or more different large language models. Then, run some experiments with the test data. For example, take some paragraphs written by Llama and ask GPT-3 to rewrite or summarize them (and vice-versa). Does your attribution model say it was written by Llama, or GPT-3? Another idea: can you engineer a prompt for GPT-3 that makes it produce text that consistently "sounds like" Llama such that it fools the attribution model (and vice-versa)? This could help test hypotheses about the features attribution models use to make their decisions — e.g. is it about word choice, sentence structures, wordiness, style, content, etc?			
22	**Use large language models for structurally focused topic modeling of natural language datasets**	Large language models can be used as powerful tools to extract structured information from natural language. Design LLM prompts to extract carefully selected types of structured information from natural language datasets (e.g., graph representations or summaries with a strictly-defined structure), and train topic models (e.g. latent Dirichlet allocation) on the extracted data to learn about themes in the data and test hypotheses (which are specific to the chosen dataset). For example, this approach was previously applied to a dataset of transcripts of YouTube videos that teach linear algebra (e.g., Khan Academy), to investigate the relationship between video popularity and the connections the instructor draws between concepts. However, the approach could be extended to natural language datasets in other domains — e.g. mental health (perhaps postings on mental-health-themed Reddit communities, or patient-therapist interactions that contain cognitive distortions), song lyrics, jokes, blog posts, scientific papers, etc.	Talbot et al 2023	8	github.com/Hramir/educational_concept_librarian
23	**Lifelong learning with latent replay in transformers**	When artificial neural networks try to learn more than one task in a sequence, they "catastrophically forget" earlier tasks. E.g., if a network trained to recognize dogs vs cats is subsequently trained to distinguish cars vs trucks, its performance on dogs vs cats drops	Ross & Andreas 2024 Jurenka et al 2024 Hadsell et al 2020	8	—

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		to chance levels very quickly. This is a fundamental problem in deep learning that has been addressed in a variety of ways — one promising group of approaches is referred to as feature-level/latent replay (see Pellegrini et al 2020). This has been successful in CNNs — in this project, try to implement a similar approach in (vision or language) transformers.	Pellegrini et al 2020 Talbot et al 2023		
24	**Neural networks as models of learners**	One goal in ML research focused on education is to develop models that can predict how students learn — for example, at what stage is each student "ready" to work on a problem of a given difficulty, and when are they vulnerable to forgetting each piece of knowledge? In this project, train a tiny CNN on a series of tasks in a continual learning setting (e.g., learn two digits from MNIST before moving on to the next two, for 5 binary classification tasks in total). Then, try to predict the tiny network's learning behavior using a more powerful model. This could be a bigger CNN that learns the same sequence of tasks and "aligns" its behavior to the smaller network, or perhaps an LSTM that makes predictions over time, etc. The ultimate goal is a model that can predict the learning behavior of a human, as opposed to a CNN model — this would be an ambitious stretch goal for a one-semester project.	—	9	—
25	**Does the effortful retrieval hypothesis translate to artificial neural networks?**	When reviewing previously-learned knowledge, humans improve the strength of their memory the most when they review something that is almost, but not quite, forgotten. The harder it is to retrieve something from memory, the more that memory is strengthened during the retrieval process — this is referred to as the "effortful retrieval hypothesis," and it is supported by a variety of behavioral studies in humans. This could be an evolutionary adaptation — maybe if an ancient human ancestor tried really hard to remember where the raspberry bushes are, that is a signal indicating that particular memory is very important, and the brain evolved some special mechanism to preserve such memories. And/or, alternatively, perhaps there is	Cognitive science references: Pyc et al 2009 Benjamin et al 2010	7	—

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		some fundamental property of neural networks such that almost-forgotten information is consolidated by “retrieval” (e.g., backpropagation in artificial neural networks) better than easily accessible information that was learned or retrieved relatively recently. Test these hypotheses using a continually-learning CNN, by systematically manipulating the timing with which each image, task, or category is presented to the network during training. Stretch goal: if there is no retrieval effort effect in CNNs, can we somehow artificially emulate one to produce a more human-like learner? (see also “Neural networks as models of learners” project above)			
26	**Detect disease in chest X-rays, and investigate how to generalize to a new clinical population**	Deep neural networks are easily confused by changes in context, which is a problem in many applications such as clinical medicine. For example, a CNN trained to interpret chest X-rays using data from the US might perform well within the US but poorly in another country. Reasons include different patient populations, different X-ray equipment, different image pre-processing, different prevalence of diseases, etc., all of which can introduce subtle shifts in the training data. This touches on issues of equity and fairness in ML, which is especially critical in clinical ML. One promising approach is to collect as diverse a dataset as possible, but in practice this can lead to a “jack of all trades, master of none” effect where the model performs decently in general but doesn't perform optimally in any given context (see Futoma et al 2020). In this project, train a CNN to detect diseases in chest X-rays, combining datasets from diverse sources in various ways. What is the best method to produce an optimal CNN that reads chest X-rays for a given clinical site (hospital, region, country, etc.), especially when data from the target site is limited? 1. Train only on the limited data from the target site?	Futoma et al 2020 Deng et al 2022	3	Kaggle: VinBigData Chest X-ray Abnormalities Detection; Kaggle: NIH Chest X-rays; StanfordML Group CheXpert; BIMCV PadChest — search the web for more!

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		2. Train on a diverse dataset from many different (other) sources and hope the model generalizes to the target site? 3. Train on a diverse dataset that includes data from the target site? 4. Train on a diverse dataset, then fine-tune to data from the target site? 5. Number 4, but mix in diverse data during the fine-tuning process? 6. In fine-tuning approaches, how much data is needed to get good results? Does the size and/or diversity of the pretraining X-ray dataset matter? In this project, pay special attention to issues of calibration and dataset balance. Consider the following: a model is trained to predict "pneumonia" vs "no pneumonia" from chest X-rays. Imagine that 2% of people getting X-rays in the US have pneumonia. The model can achieve 98% accuracy by ALWAYS guessing "no pneumonia." But maybe in Australia, for some reason 5% of people getting X-rayed have pneumonia — now the same model only has 95% accuracy. Even a model that is doing more than just always guessing no-pneumonia will tend to be "calibrated" to the base disease rate of the population it was trained on, which might be different from the target population.			
27	**Humanity's last exam**	This project follows on an ongoing challenge (agi.safe.ai/submit). Can you think about questions and problems that cannot be solved by current AI? The idea is to systematically develop the themes/questions/topics/problem sets that cannot be solved by current AI. This should be in the form of vision/language challenges. For example, "AIs cannot play soccer like Lionel Messi" is not compliant with this challenge. What families of math/history/literature/vision/physics/reasoning/legal/clinical, etc. challenges cannot be solved by current AIs?	—	3	agi.safe.ai/submit